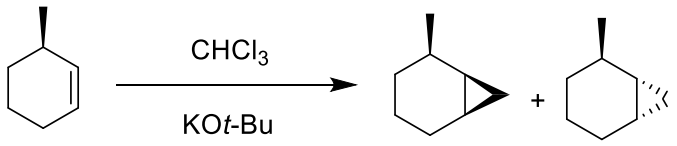
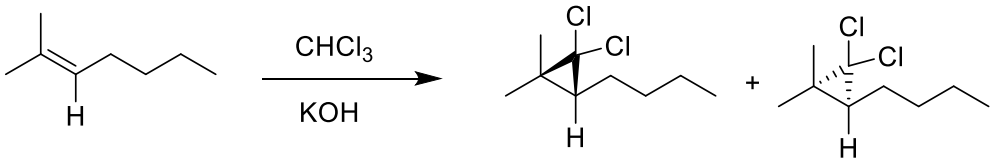
1.

a.



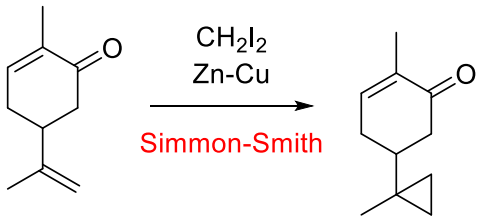
mixture of diastereomers

b.



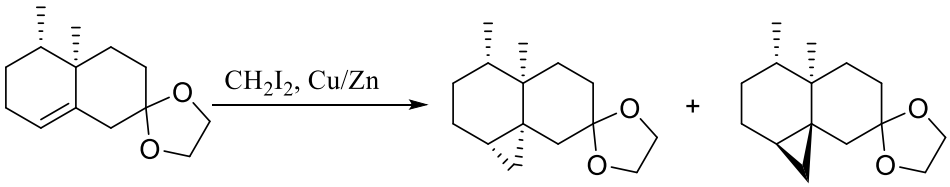
mixture of enantiomers

c.

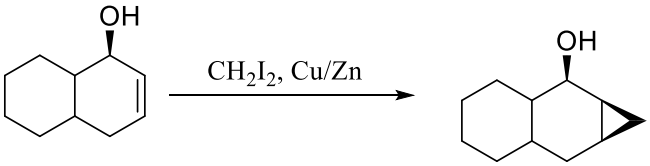


Regioselective ⟹ electron-rich double-bond reacts faster

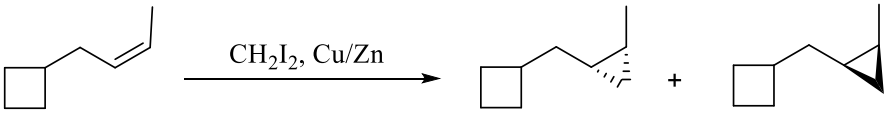
d.



e.



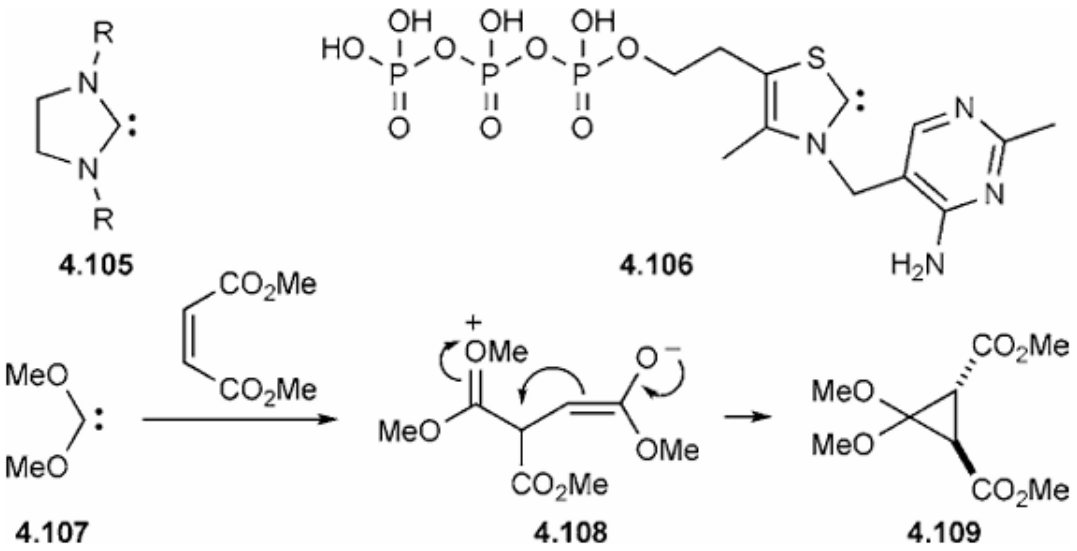
f.



2. Yes, Carbene can act as electrophiles as well as nucleophiles

4.6.1 Nucleophilic Carbenes

In practice, donor substituents make it possible actually to isolate a range of carbenes **4.105**. With somewhat less stabilisation, the carbene **4.106**, although it is only found as a reactive intermediate, is exceptionally easy to form. It is the key intermediate in all the metabolic steps catalysed by thiamine coenzymes, and its reactions are characterised by its nucleophilicity. Similarly, dimethoxycarbene **4.107** reacts as a nucleophile with electrophiles like dimethyl maleate to give the intermediate **4.108**, and hence the cyclopropane **4.109**, but it does not insert into unactivated alkenes.



The well known cycloaddition of a carbene to an alkene, to be discussed again in the next chapter, can be viewed as the simultaneous interaction of the HOMO of the alkene with the LUMO of the carbene, and of the LUMO of the alkene and the HOMO of the carbene. Nucleophilic carbenes with a high-energy HOMO, interact strongly with molecules having a low-energy LUMO (Fig. 4.14a), which is why they react well with electrophilic alkenes. In the case of the very nucleophilic dimethoxycarbene, bond formation is so entirely dominated by the $HOMO_{(carbene)}$-$LUMO_{(alkene)}$ interaction that it gives the zwitterionic intermediate **4.108** rather than undergoing a concerted cycloaddition.

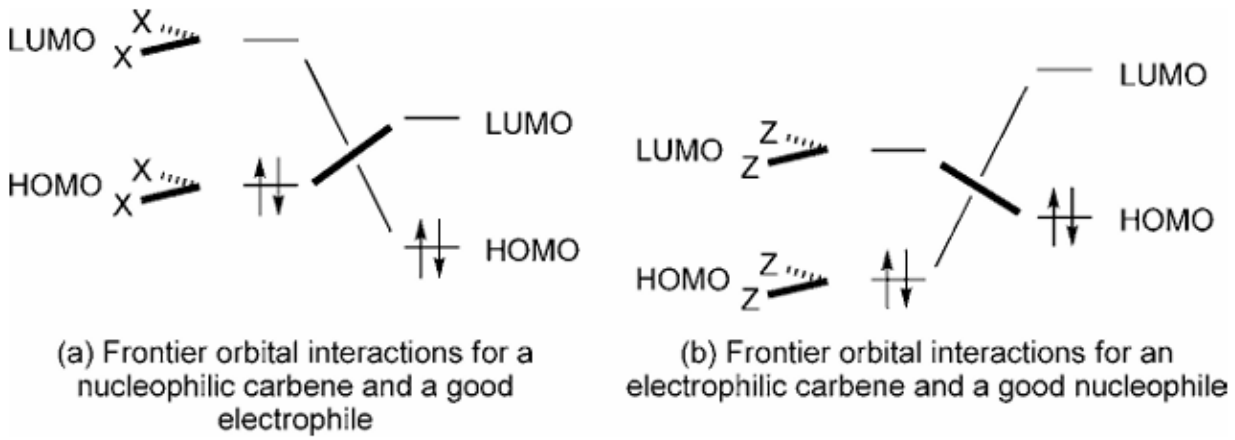
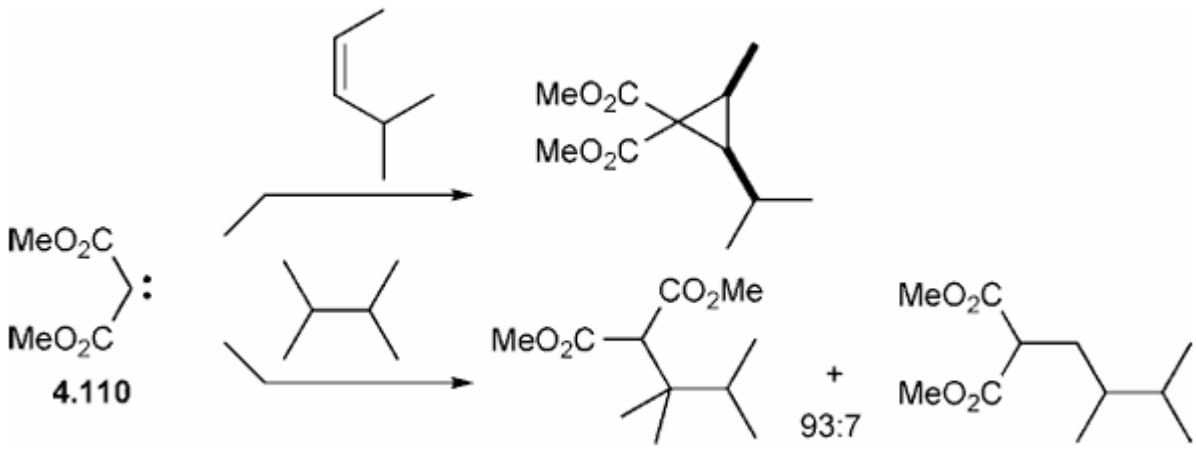


Fig. 4.14 Frontier orbital interactions for carbenes with electrophilic and nucleophilic reagents

4.6.2 Electrophilic Carbenes

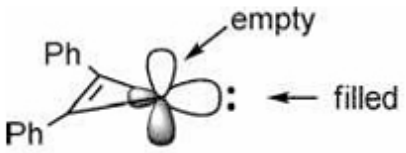
Electrophilic carbenes, however, like the bis(methoxycarbonyl)carbene **4.110**, have a low-energy LUMO, and react with molecules like alkenes that have a high-energy HOMO (Fig. 4.14b), and cyclo-add stereospecifically. They also insert into C—H bonds, with high selectivity for tertiary C—H bonds, which argues for a substantial degree of cationic charge on the carbon in the transition structure, characteristic of electrophilic attack on the hydrogen atom. Dihalocarbenes are electrophilic in character, inserting easily and stereospecifically syn into C=C bonds, but not usually reacting with C—H bonds.



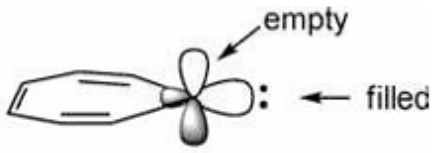
3. EPR active = $:CH_2$, $:CMe_2$ triplet carbens
   EPR inactive = $:CCl_2$, $:C(OMe)_2$ singlet carbens

4.6.3 Aromatic Carbenes

The carbenes cyclopropenylidene **4.111** and cycloheptatrienylidene **4.112** have the empty p orbital conjugated with one and three π bonds, respectively, making them aromatic like the cyclopropenyl **1.10** and tropylium cations **1.9**. The filled p_x orbital is unchanged as a source of nucleophilicity, and these carbenes are notably nucleophilic, reacting with electrophilic alkenes like fumarate, but not with simple alkenes.



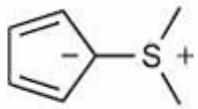
4.111



4.112

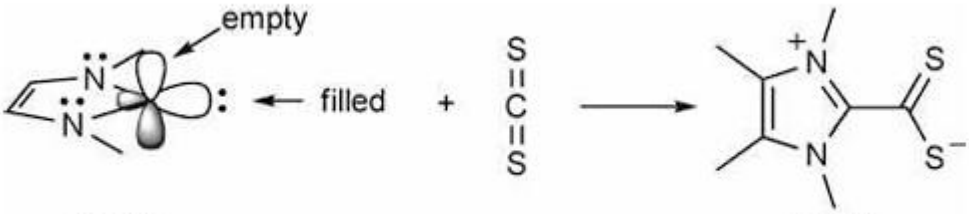
Cyclopentadienylidene is not so straightforward. It might change the normal configuration for a carbene **4.113** to that shown as **4.114** in order to allow the filled p_z orbital to be conjugated with the two π bonds like the cyclopentadienyl anion **1.8**, and the unfilled p_x orbital would have to take up the orthogonal role. However, this is not without an energetic penalty, since it keeps the areas of high electron population close together on the left-hand side of **4.114**. This carbene is not notably electrophilic in its reactions with alkenes, but it is somewhat electrophilic, reacting with dimethyl sulfide, for example, to give the ylid **4.115**.

4.113 4.114



4.115

The superficially similar carbene **4.116**, another carbene stable enough to be isolated, has the best of all worlds. With six electrons for the π system coming from the double bond and the two nitrogen lone pairs, it has an aromatic sextet without having to fill the p_z orbital. The p_z orbital is empty and the p_x orbital is filled, making this a nucleophilic carbene, which reacts with the electrophile carbon disulfide to give the zwitterion **4.117**.



4.116 4.117